SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5116
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage: **40%**

QUESTIONS:

Total Marks: 50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Vengati Mude Kavitha (192312567) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n $\phi(n)$\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i. Calculate total encryption time for each algorithm
ii. Analyze trade-offs between security and performance
iii. Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

: CO2-AT1 :-

Block Cipher Mode Evaluation :-

Given,

$P = [1010, 1010, 1111, 1010]$

$K = 1100$

Initialization Vector (IV) : 0000

Assumption :- Since the question does not specify a block cipher function, use a simplified XOR-based cipher for illustration

$$E_K(X) = X \oplus K.$$

a) ECB Mode Encryption,

In ECB mode, each plaintext block is encrypted independently.

Formula, $C_i = P_i \oplus K.$

Calculations :-

$C_1 = 1010 \oplus 1100 = 0110$

$C_2 = 1010 \oplus 1100 = 0110$

$C_3 = 1111 \oplus 1100 = 0011$

$C_4 = 1010 \oplus 1100 = 0110$

ECB Cipher text = [0110, 0110, 0011, 0110]

b) CBC Mode Encryption,

In CBC mode, each plaintext block is XORed with the previous ciphertext block before encryption.

Formula, $C_i = E_K(P_i \oplus C_{i-1})$.

Initial Value:- $C_0 = IV = 0000$

$C_1 = (1010 \oplus 0000) \oplus 1100 = 0110$

$C_2 = (1010 \oplus 0110) \oplus 1100 = 0000$

$C_3 = (1111 \oplus 0000) \oplus 1100 = 0011$

$C_4 = (1010 \oplus 0011) \oplus 1100 = 0101$

Ciphertext :- [0110, 0000, 0011, 0101]

c, Comparison of Ciphertext Outputs:-

Feature	ECB Mode	CBC Mode
Encryption	Independent blocks	Blocks are chained
Identical plaintext blocks	Produce identical ciphertext	Can produce different ciphertext
pattern hiding	Poorer	Better.
Error effect	Limited to a block	Can affect multiple blocks

d, Analysis & Conclusion:-

ECB produces the same ciphertext for identical plaintext blocks, so repeated patterns remain visible. CBC uses the previous ciphertext block, making encryption dependent on earlier blocks & helping hide repeated plaintext patterns.

ECC vs RSA Comparison

Given,

RSA = 1024 bits

ECC = 160 bits

Device = IoT device with limited processing capacity

1, Computational Cost:-

RSA uses larger keys & requires large-integer arithmetic especially during key generation & private key operations. ECC uses smaller keys but involves elliptic-curve point multiplication.

For equivalent security levels, ECC generally requires less computation & communication overhead than RSA, particularly for key storage & transmission.

2, Power & Memory Consumption:-

- RSA:- Requires larger key storage & more memory for large-number calculations. It generally has higher communication overhead.
- ECC:- Uses smaller keys, requires less memory for key storage, & reduces the amount of data transmitted. It is generally more energy-efficient for constrained IoT devices.

3, Comparison Table:-

Parameter	RSA	ECC.
Key Size	1024 bits	160 bits
Memory usage	Higher	Lower.
Communication overhead.	Higher	Lower
Processing	Large-integer arithmetic	Elliptic curve operations

4, Conclusion:-

ECC is better choice for the given IoT Scenario because it provides comparable security with smaller keys, lower storage requirements, and reduced Communication overhead. However, the given 160 bit ECC key and 1024-bit RSA key are historical sizes. and should not be treated as equivalent security choices for modern deployments.

5. ECC vs RSA Comparison

Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation